WORKED SOLUTION

# STEP II 2014, Question 12

Probability and Statistics • Hazard Functions

**STEP II 2014, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q12

## The key idea

The hazard function is the rate of dying *for those still alive*. Everything in the later parts follows from one observation: since $F'(t) = f(t)$,

$$h(t) = \frac{f(t)}{1 - F(t)} = -\frac{\mathrm{d}}{\mathrm{d}t} \ln(1 - F(t)).$$

So knowing $h$ means knowing the derivative of $\ln(1 - F)$. Integrate, use the fact that $F = 0$ where the lifetime starts, and differentiate $F$ to get $f$.

## Solution

### Part (i)

Let $A$ be the event $T \geq t$ (the fly is alive at time $t$) and $B$ the event $t \leq T \leq t + \delta t$ (it dies in the next $\delta t$). Since $B \cap A = B$ (for a continuous variable $\mathrm{P}(T = t) = 0$, so it doesn't matter whether the endpoints are included),

$$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(B)}{\mathrm{P}(A)} = \frac{F(t + \delta t) - F(t)}{1 - F(t)}.$$

For small $\delta t$, $F(t + \delta t) - F(t) \approx F'(t)\,\delta t = f(t)\,\delta t$, so

$$\mathrm{P}(B \mid A) \approx \frac{f(t)\,\delta t}{1 - F(t)} = h(t)\,\delta t,$$

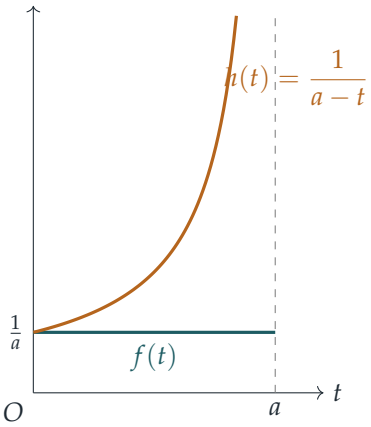
as required. (We need $F(t) < 1$ so that we are not conditioning on an impossible event; this is exactly where $h$ is defined.)

### Part (ii)

Here $f(t) = F'(t) = \frac{1}{a}$ for $0 < t < a$: $T$ is uniform on $(0, a)$. Then

$$h(t) = \frac{1/a}{1 - t/a} = \frac{1}{a - t}, \qquad 0 < t < a.$$

The density is flat, but the hazard starts at $h(0) = \frac{1}{a}$ and increases, with $h(t) \to \infty$ as $t \to a^-$. That makes sense: a fly that has survived to almost time $a$ is certain to die in the tiny time remaining.




(The diagram is drawn with $a = 2$.)

**Part (iii)**

Using the key idea, $-\dfrac{\mathrm{d}}{\mathrm{d}t}\ln(1-F(t)) = \dfrac{1}{t}$ for $t > a$. Integrating from $a$ to $t$, and using $F(a) = 0$ because $T$ takes values only above $a$,

$$-\ln(1-F(t)) = \ln t - \ln a = \ln\frac{t}{a}, \qquad \text{so} \qquad 1 - F(t) = \frac{a}{t}.$$

Hence $F(t) = 1 - \dfrac{a}{t}$ and

$$f(t) = \frac{a}{t^2}, \qquad t > a$$

(and $f(t) = 0$ for $t \le a$). As a check, $\int_a^\infty at^{-2}\,\mathrm{d}t = 1$, and $f/(1-F) = (a/t^2)/(a/t) = 1/t$.

**Part (iv)**

**"If".** Suppose $f(t) = k\mathrm{e}^{-k(t-b)}$ for $t > b$ with $k > 0$. Since $\int_b^\infty k\mathrm{e}^{-k(t-b)}\,\mathrm{d}t = 1$, all the probability lies in $t > b$, so $f(t) = 0$ and $F(t) = 0$ for $t \le b$. Hence $h(t) = 0/1 = 0$ for $t \le b$. For $t > b$,

$$F(t) = \int_b^t k\mathrm{e}^{-k(s-b)}\,\mathrm{d}s = 1 - \mathrm{e}^{-k(t-b)}, \qquad h(t) = \frac{k\mathrm{e}^{-k(t-b)}}{\mathrm{e}^{-k(t-b)}} = k,$$

which is constant.

**"Only if".** Suppose $h(t) = 0$ for $t \le b$ and $h(t) = k$ (a constant) for $t > b$.

**Ask yourself** What does $h(t) = 0$ for $t \leq b$ tell us about $F(b)$?

Where $h(t) = 0$ we have $f(t) = h(t)(1 - F(t)) = 0$. So $f$ vanishes for all $t \leq b$, giving $F(t) = \int_{-\infty}^{t} f = 0$ for $t \leq b$; in particular $F(b) = 0$. (This also confirms $F < 1$ there, so $h$ really is defined on $t \leq b$.)

For $t > b$, $-\dfrac{\mathrm{d}}{\mathrm{d}t}\ln(1 - F(t)) = k$. Integrating from $b$ to $t$ and using $F(b) = 0$,

$$-\ln(1 - F(t)) = k(t - b), \qquad F(t) = 1 - \mathrm{e}^{-k(t-b)},$$

and differentiating, $f(t) = k\mathrm{e}^{-k(t-b)}$ for $t > b$. Finally, $k$ must be positive: $k \geq 0$ because $f \geq 0$, and if $k = 0$ then $F$ would stay at 0 for ever, so $F(t) \not\to 1$ and $T$ would not be a genuine random variable. This completes both directions.

**Part (v)**

Again $-\dfrac{\mathrm{d}}{\mathrm{d}t}\ln(1 - F(t)) = \dfrac{\lambda}{\theta^{\lambda}}t^{\lambda-1}$ for $t > 0$. Integrating from 0 to $t$, with $F(0) = 0$ since $T > 0$, and noting that $\lambda > 0$ makes $t^{\lambda} \to 0$ as $t \to 0$,

$$-\ln(1 - F(t)) = \frac{\lambda}{\theta^{\lambda}} \cdot \frac{t^{\lambda}}{\lambda} = \left(\frac{t}{\theta}\right)^{\lambda}.$$

So $F(t) = 1 - \exp(-(t/\theta)^{\lambda})$, and differentiating,

$$f(t) = \frac{\lambda}{\theta^{\lambda}}\, t^{\lambda-1} \exp\!\left(-\left(\frac{t}{\theta}\right)^{\lambda}\right), \qquad t > 0.$$

This is a genuine density: it is positive, and $F(t) \to 1$ as $t \to \infty$.

## Checks

- Each answer can be checked by computing $f/(1 - F)$ directly. For example, in (v) the exponential factors cancel and leave $\lambda\theta^{-\lambda}t^{\lambda-1}$.
- Part (v) with $\lambda = 1$ gives constant hazard $1/\theta$ and $f(t) = \frac{1}{\theta}\mathrm{e}^{-t/\theta}$, which is part (iv) with $b = 0$, $k = 1/\theta$.
- Numerically, with $\lambda = \frac{1}{2}$, $\theta = 2$, integrating $f$ over $(0, \infty)$ gives 1.0000, and $F(1) = 1 - \mathrm{e}^{-1/\sqrt{2}} \approx 0.5069$ agrees with direct integration of $f$ from 0 to 1.

## Common slips

- Forgetting the condition $F = 0$ at the start of the lifetime, and leaving an arbitrary constant of integration in $F$.
- In (iv), proving only one direction of the "if and only if", or not using $h = 0$ for $t \leq b$ to deduce $F(b) = 0$.
- Losing the minus sign: $\frac{\mathrm{d}}{\mathrm{d}t}\ln(1 - F) = -f/(1 - F)$.

## Beyond STEP

The relation $1 - F(t) = \exp\left(-\int_0^t h(s)\,\mathrm{d}s\right)$ is the foundation of *survival analysis* and reliability theory. The distribution in (v) is the Weibull distribution: $\lambda < 1$ gives a falling hazard (early failures), $\lambda = 1$ the memoryless exponential, and $\lambda > 1$ a rising hazard (wearing out). Survival analysis is used everywhere from medical trials to engineering. For the underlying probability, see Blitzstein and Hwang, *Introduction to Probability*.

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